

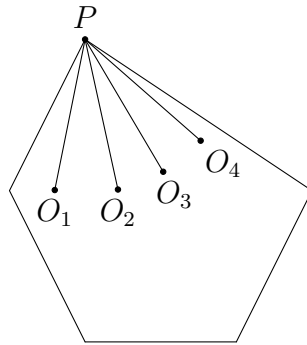
Combinatorial Geometry — Solutions

Problem 1. (JBMO Shortlist 2015 C2) 2015 points are given in a plane such that from any five points we can choose two points with distance less than 1 unit. Prove that 504 of the given points lie on a unit disc.

Solution. Let A, B, C, D be any four points among the 2015 points. Consider the four unit discs with A, B, C, D as centres. Among the remaining 2011 points, at least $\left\lceil \frac{2011}{4} \right\rceil = 503$ of them lie in the same unit disc by the pigeonhole principle (it doesn't matter that each point may belong to more than one unit disc). Together with the centre, these 504 points lie in the same unit disc. $\square$

Problem 2. (Yugoslavia TST 1996 Problem 2) Let there be given a set of 1996 equal circles in the plane, no two of them having common interior points. Prove that there exists a circle touching at most three other circles.

Solution. Consider the set S containing all centres of the circles. If all points in S are collinear, it is clear that each circle is tangent to at most two other circles, and we are done. Otherwise, let P be a vertex of the convex hull of S . We claim that the circle Γ with centre P is tangent to at most three other circles.



Suppose on the contrary that Γ is tangent to four circles with centres O_1, O_2, O_3, O_4 respectively. Then $PO_1 = PO_2 = PO_3 = PO_4 = 2R$, where R is the radius of each circle. WLOG assume PO_1, PO_2, PO_3, PO_4 lie in this order as shown. Since P is a vertex of the convex hull, we have $\angle O_1PO_4 < 180^\circ$. By the pigeonhole principle, one of $\angle O_1PO_2, \angle O_2PO_3$ and $\angle O_3PO_4$ is less than 60° . WLOG assume $\angle O_1PO_2 < 60^\circ$. Since $PO_1 = PO_2$, this implies $O_1O_2 < PO_1 = 2R$. But then the circles with centres O_1 and O_2 intersect, which is a contradiction. So the claim holds, and the proof is complete. $\square$

Problem 3. There are k points inside a convex polygon $\mathcal{P}$ with n vertices, where $2 \leq k \leq \left\lfloor \frac{n}{2} \right\rfloor$. Prove that we can choose $2k$ vertices of $\mathcal{P}$ such that the polygon formed by these vertices contains all the k points.

Solution. Initially, let $\mathcal{S}$ be the set of all vertices of $\mathcal{P}$. We remove some vertices from $\mathcal{S}$ as follows. Suppose $|\mathcal{S}| = m > 2k$. Let $A_1 A_2 \cdots A_m$ be the polygon with vertices in $\mathcal{S}$. Consider the following triangles.

$$\triangle A_1 A_2 A_3, \triangle A_2 A_3 A_4, \dots, \triangle A_m A_1 A_2$$

Note that each of the k points lies in at most two of these triangles, since only consecutive triangles in this list overlap (including the first and the last), and $m > 2k \geq 4$. As $m > 2k$, there must be a triangle not containing any of the k points. WLOG assume $\triangle A_1 A_2 A_3$ is such a triangle. Then we remove A_2 from $\mathcal{S}$ so that all k points still lie in the convex hull of $\mathcal{S}$.

We repeat the same process until $|\mathcal{S}| = 2k$. This is possible since initially we have $|\mathcal{S}| = n \geq 2k$. Clearly, we can just take the $2k$ vertices in $\mathcal{S}$ at the end. $\square$

Problem 4. (Dutch TST 2017 Test 3 Problem 4) Let $n \geq 2$ be an integer. Find the smallest positive integer m for which the following holds: given n points in the plane, no three on a line, there are m lines such that no line passes through any of the given points, and for all points $X \neq Y$ there is a line with respect to which X and Y lie on opposite sides.

Solution. The answer is $\left\lceil \frac{n}{2} \right\rceil$.

Firstly, we prove that $m \geq \left\lceil \frac{n}{2} \right\rceil$. Consider the case when the n points lie on the same circle. Let $P_1, P_2, \dots, P_n$ be these points in clockwise direction. In order that P_1, P_2 are separated by a line, there must be a line intersecting the arc $P_1 P_2$ not containing the other points. Similarly, there is a line intersecting each of the arcs $P_2 P_3, \dots, P_{n-1} P_n, P_n P_1$. As each line intersects at most two arcs, we must have $m \geq \left\lceil \frac{n}{2} \right\rceil$.

Secondly, we prove that $m \leq \left\lceil \frac{n}{2} \right\rceil$. Indeed, given any n points, we can construct $k^+ = \left\lceil \frac{n}{2} \right\rceil$ lines that meet the condition. Consider a line ℓ not parallel to any line formed by joining two of the points such that all points lie on the same side of ℓ . WLOG assume ℓ is horizontal, and all points lie above ℓ . We move ℓ upwards gradually until it passes through k^+ points (this is possible since each time it passes through exactly one point). At this stage, there are k^+ points below ℓ and $k^- = \left\lfloor \frac{n}{2} \right\rfloor$ points above ℓ .

For $j = 1, 2, \dots, k^+ - 1$, we choose two points A, B above ℓ which are not separated by any existing line, and choose two points C, D below ℓ which are not separated by any existing line. If there is no pair of points on one side of ℓ which are not separated, we can just pick any pair of points on that side. Since no three points are collinear, we can find a line ℓ_j cutting the segments AB and CD which does not pass through all the n points. Now, as each new line added separates at least one more pair of points below ℓ (unless all pairs are already separated), all k^+ points below ℓ are separated.

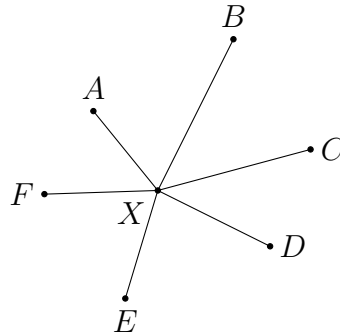
Similarly, all k^- points above ℓ are separated. Also, the points above ℓ are clearly separated from the points below ℓ by ℓ . As there are k^+ lines in total, we are done. $\square$

Problem 5. (Taiwan TST 2021 Round 3 Quiz 1 Problem C) A city is a point on the plane. Suppose there are $n \geq 2$ cities. Suppose that for each city X , there is another city $N(X)$ that is strictly closer to X than all the other cities. The government builds a road connecting each city X and its $N(X)$; no other roads have been built. Suppose we know that, starting from any city, we can reach any other city through a series of road.

We call a city Y *suburban* if it is $N(X)$ for some city X . Show that there are at least $\frac{n-2}{4}$ suburban cities.

Solution. Consider the graph whose vertices are the n cities such that an edge is drawn between every two vertices X and $N(X)$. Note that if Y and Z are two closest cities, then $N(Y) = Z$ and $N(Z) = Y$. Therefore, there are at most $n - 1$ edges in this graph. On the other hand, it is given that the graph is connected. Thus, it contains exactly $n - 1$ edges (and it is a tree). Also, this implies Y and Z are the only pair of cities which are the closest cities of each other.

We claim that the degree of every vertex is at most 5. Suppose on the contrary that $\deg X \geq 6$ for some X . Let A, B, C, D, E, F be adjacent vertices to X , such that XA, XB, XC, XD, XE, XF lie in this order as shown. WLOG assume $\angle AXB \leq 60^\circ$ by the pigeonhole principle. Then we may assume XB is the longest side in $\triangle XAB$. But this implies $N(X) \neq B$ and $N(B) \neq X$, contradicting the fact that there is an edge between B and X . Thus, the claim holds.



Now, suppose there are k suburban cities. For each of the $k - 2$ suburban cities X different from Y and Z , as the degree of X is at most 5 and one edge incident on X joins X and $N(X)$ (while $N(N(X)) \neq X$), X is the closest city of at most 4 cities. For each of Y and Z , it is the closest city of at most 5 cities. Since each city has exactly one closest city, we have

$$n \leq 4(k - 2) + 5(2) = 4k + 2.$$

It follows that $k \geq \frac{n-2}{4}$. $\square$

Problem 6. Let $\mathcal{S}$ be an infinite set of points on the plane. Suppose for any $A, B, C \in \mathcal{S}$, the distance from A to BC is an integer. Prove that all the points in $\mathcal{S}$ are collinear.

Solution. Suppose on the contrary that not all points in $\mathcal{S}$ are collinear. We fix two points A and B in $\mathcal{S}$. We first prove that there is only a finite number of points not on AB .

For each point $C \in \mathcal{S}$ not lying on AB , the distance from B to AC is $AB \sin \angle BAC$. As this is an integer lying between 1 and AB , it has finitely many different possibilities. Clearly, each possibility fixes $\angle BAC$, and each angle corresponds to at most two directions of AC . This shows C must lie on a finite number of lines $\ell_1, \ell_2, \dots, \ell_k$ (different from AB) passing through A . Similarly, C lies on a finite number of lines $L_1, L_2, \dots, L_K$ passing through B . As each pair of ℓ_i and L_j has at most one intersection point, there are only finitely many such points C which can lie on some ℓ_i and some L_j simultaneously. This proves our claim.

Now, as $\mathcal{S}$ is infinite, there are infinitely many points on AB . Let C be any point not on AB . Applying the claim to the line AC , we see that there are finitely many points not on AC . But then every point on AB other than A does not lie on AC . This is a contradiction. Therefore, all points in $\mathcal{S}$ are collinear. $\square$

Problem 7. (Canada MO 2017 Problem 5) There are 100 circles of radius one in the plane. A triangle formed by the centres of any three given circles has area at most 2017. Prove that there is a line intersecting at least three of the circles.

Solution. Suppose on the contrary that every line intersects at most two circles. Let A and B be the centres of two circles such that AB is maximized. We set up a coordinate system such that AB is the x -axis. Let the centres of the circles be $P_j(x_j, y_j)$ where $j = 1, 2, \dots, 100$ and $x_1 \leq x_2 \leq \dots \leq x_{100}$. Since $P_1 P_{100} \geq x_{100} - x_1 \geq AB$, equality must hold by the choice of A, B . Thus, we have $P_1 = A$ and $P_{100} = B$.

We claim that $AB \geq 98$. Indeed, if $AB < 98$, by partitioning $[x_1, x_{100}]$ into 49 intervals of equal length, the length of each interval is less than 2. By the pigeonhole principle, there exist x_j, x_{j+1}, x_{j+2} lying in the same interval. Then the straight line $x = \frac{x_j + x_{j+2}}{2}$ intersects the three circles with centres P_j, P_{j+1}, P_{j+2} , since $\left| \frac{x_j + x_{j+2}}{2} - x_i \right| < 1$ for $i = j, j+1, j+2$. This is a contradiction. So $AB \geq 98$.

Similarly, if we arrange the y -coordinates of the points in the order $y_{i_1} \leq y_{i_2} \leq \dots \leq y_{i_{100}}$, we must have $y_{i_{100}} - y_{i_1} \geq 98$. Since A, B have the same y -coordinate, there exists a circle with centre C (with y -coordinate y_{i_1} or $y_{i_{100}}$) which has a distance ≥ 49 from AB . This gives

$$[ABC] \geq \frac{98 \times 49}{2} = 2401 > 2017,$$

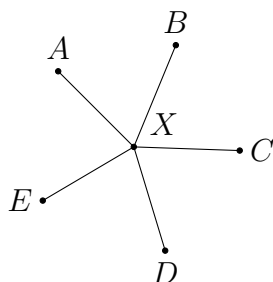
contradicting the assumption. Thus, there is always a line intersecting at least three circles. $\square$

Problem 8. (Korea TST Final Round 2018 Problem 6) Twenty ants live on the faces of an icosahedron, one ant on each face, where the icosahedron has each side with length 1. Each ant moves in a counterclockwise direction on each face, along the edges. The speed of each ant must be no less than 1 always. Also, if two ants meet, they should meet at the vertex of the icosahedron. If five ants meet at the same time at a vertex, we call that a *collision*. Can the ants move forever, in a way that no collision occurs?

Solution. No. We transform the problem step by step.

Claim 1. If it is possible that no collision occurs, then it is possible that all the ants never meet each other and the speed of each ant is always at least 0.9.

Proof. Consider a movement such that no collision occurs. Suppose there is a moment when at least 2 (and at most 4) ants meet at a vertex X . Let XA, XB, XC, XD, XE be edges of the icosahedron that lie in this order clockwise around X . Just before the ants meet, they are moving towards X on some of these edges. WLOG assume no ant moves from E to X , and there is an ant moving from D to X . We may assume this ant speeds up a little bit when it is very close to X , so that it passes through the vertex X without meeting the other ants. Afterwards, this ant can slow down a little bit to restore the original trajectory. Next, if there is another ant moving from C to X , it can speed up a little bit so that it is the second one to pass through X , etc. (If there is no such ant, we just consider the next edge.) As a result, it is possible to make the ants move without meeting each other. By doing this for every such moment, we have proven claim 1.

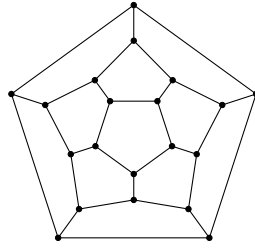


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At any moment, we can associate an edge to each ant such that the ant is moving on this edge. When an ant passes through a vertex, it changes from one edge to another. By claim 1, each time only one ant changes its associated edge. Also, note that no two ants move on the same edge at any moment, since otherwise they must be moving in opposite directions on this edge, and hence they must meet somewhere on this edge, which contradicts the assumption. We further remove the speed constraint by imposing a much weaker condition, that is, every ant must change its associated edge infinitely often. Clearly, this can be deduced from the speed constraint. Now, we only need to know which edges the ants lie on at every moment.

We further transform the problem using graph theory. Consider a directed graph as follows. The vertices of this graph are the 20 faces of the icosahedron. We draw a directed edge from V to W if the ant on face V is moving on the common edge of the faces V and W . Note that the graph changes whenever an ant changes its edge. We have the following observations.

- There is exactly one edge from each vertex.
- If there is an edge from V to W , then there is no edge from W to V .
- If we ignore the directions of the edges, then the graph is a subgraph of the planar graph as shown (which is the Schlegel diagram of a dodecahedron, the dual of an icosahedron).



- Every time when an ant passes through a vertex, one of the directed edges VW changes to another directed edge VX which is the next edge from V in the anticlockwise direction. Also, the edge from each vertex changes infinitely often.

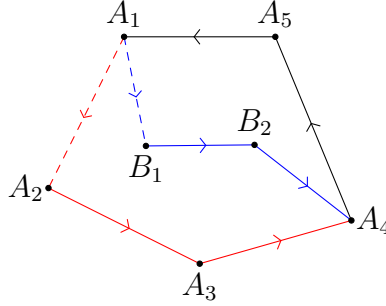
We are going to show that it is impossible for this to occur for any finite planar graph.

Claim 2. Let G be a finite simple planar graph. Consider a directed graph consisting of all vertices of G such that there is exactly one edge from each vertex (which is an edge of G with direction), and no two edges are opposite of each other. Each time we change a directed edge VW to the next edge VX in the anticlockwise direction, so that the conditions are still satisfied. Then it is impossible that the edge from each vertex changes infinitely often.

Proof. Suppose on the contrary that the edge from each vertex changes infinitely often. At every moment, we partition all vertices as follows. For each vertex A_1 , we can follow the directed edges to form the walk $A_1 \rightarrow A_2 \rightarrow \dots$ in a unique way. Since there are finitely many vertices, we must reach a vertex A_k which is the same as a previous vertex A_j . Then $A_j \rightarrow A_{j+1} \rightarrow \dots \rightarrow A_{k-1} \rightarrow A_j$ is a cycle. We put A_1 in the same group as all vertices in this cycle. Clearly, no two cycles share common vertices, and each vertex corresponds to exactly one cycle. Therefore, all vertices are partitioned into several groups, where each group consists of one cycle.

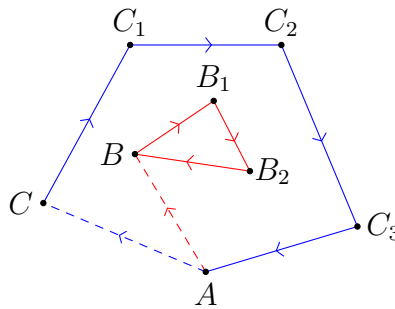
Suppose there is a cycle $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_k \rightarrow A_1$ in the anticlockwise direction. We claim that there is such a cycle enclosing a region with smaller and smaller area as time goes by. WLOG assume the edge A_1A_2 changes to some A_1B_1 after some time. Due to the orientation, B_1 lies inside or on the boundary of the region enclosed by $A_1A_2 \dots A_k$. We continue to move along the directed edges. Since the graph is planar, we must eventually go into a cycle lying strictly inside $A_1A_2 \dots A_k$, or reach one of the vertices $A_1, A_2, \dots, A_k$. In the former case, we already get a cycle of smaller area. In the latter case, we can continue to move along the boundary to go back to A_1 . Again, this new cycle lies strictly inside $A_1A_2 \dots A_k$, and hence it has smaller area. However, since there is only

a finite number of cycles, it is impossible that the area keeps on decreasing. (In fact, when the cycle is a triangle, the three edges can no longer change.)



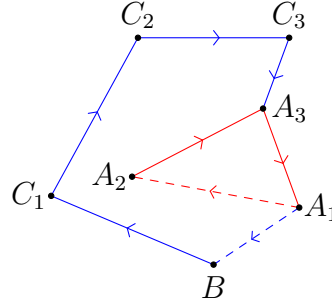
Therefore, from now on we assume all cycles are clockwise. Note that if the regions enclosed by two cycles overlap, one of them must lie strictly inside another. We say that a cycle is *maximal* if it does not lie inside any other cycle. Let N be the number of maximal cycles. We show that N is monotonic decreasing, and when it becomes a constant (which must happen as there are finitely many possibilities of N), the total area S of the regions enclosed by the maximal cycles keeps on increasing. There are a few possibilities when an edge changes.

- Suppose the edge AB not in a cycle changes to AC , where C is not in the same group as A . This does not break any existing cycle or form any new cycle. So N and S do not change.
- Suppose the edge AB not in a cycle changes to AC , where C is in the same group as A . All existing cycles remain unchanged. The only possibility for a new cycle to form is that there exists a path $C \rightarrow C_1 \rightarrow \dots \rightarrow C_k \rightarrow A$. As we have assumed $ACC_1 \dots C_k$ is clockwise, and AB changes to AC in the anticlockwise direction, B must lie inside $ACC_1 \dots C_k$. Then the original cycle in the same group as A lies strictly inside $ACC_1 \dots C_k$, and so N remains unchanged, while S does not decrease (it may not increase as $ACC_1 \dots C_k$ may not be maximal).



- Suppose the edge A_1A_2 in a cycle $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_k \rightarrow A_1$ changes to A_1B , where B is not in the same group as A_1 . Starting from any vertex which is in the same group as A_1 , we can eventually reach A_1 . Then we move to B , and eventually enter its cycle. So one cycle disappears and the other cycles are unaffected. The values of N and S do not increase. Note that N decreases if $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_k \rightarrow A_1$ is maximal.

- Suppose the edge A_1A_2 in a cycle $A_1 \rightarrow A_2 \rightarrow \cdots \rightarrow A_k \rightarrow A_1$ changes to A_1B , where B is in the same group as A_1 . Since $A_1A_2 \cdots A_k$ is clockwise, B must lie outside $A_1A_2 \cdots A_k$. As B is in the same group as A_1 , we must eventually go back to some A_j , and hence A_1 . This forms a new cycle which is strictly larger than the original cycle $A_1 \rightarrow A_2 \rightarrow \cdots \rightarrow A_k \rightarrow A_1$. Thus, N remains unchanged, while S does not decrease. Note that S increases if the cycle $A_1 \rightarrow A_2 \rightarrow \cdots \rightarrow A_k \rightarrow A_1$ is maximal.



This shows N is monotonic decreasing, and eventually it becomes a constant. Afterwards, every time when an edge in a maximal cycle changes, the value of S increases. However, it is obvious that S cannot increase indefinitely. This is a contradiction. So we have proven claim 2. $\square$

As a corollary, we have solved the original problem. $\square$